2.29 Integration: Advanced Methods (Trigonometric)

Worked Solutions

1 a)

Let $x = \tan u$, so $dx = \sec^2 u\, du$.

$$\begin{aligned}\int \frac{x^2}{1+x^2}\, dx &= \int \frac{\tan^2 u}{1+\tan^2 u} \sec^2 u\, du = \int \tan^2 u\, du \\ &= \int (\sec^2 u - 1)\, du = \tan u - u + c = x - \arctan x + c.\end{aligned}$$

1 b)

$$\begin{aligned}\frac{x^2}{1+x^2} &\equiv \frac{x^2+1-1}{1+x^2} = \frac{x^2+1}{1+x^2} - \frac{1}{1+x^2} \\ &\equiv \frac{1+x^2}{1+x^2} - \frac{1}{1+x^2} = 1 - \frac{1}{1+x^2}.\end{aligned}$$

Or using polynomial division, the quotient is 1 and the remainder is -1, so $\frac{x^2}{1+x^2} \equiv 1 - \frac{1}{1+x^2}$. Hence

$$\int \frac{x^2}{1+x^2}\, dx = \int \left(1 - \frac{1}{1+x^2}\right) dx = x - \arctan x + c.$$

2 a)

Let $x = \tan u$, so $dx = \sec^2 u\, du$.

$$\begin{aligned}\int \frac{2x^2-1}{1+x^2}\, dx &= \int (2\tan^2 u - 1)\, du = \int \big(2(\sec^2 u - 1) - 1\big)\, du \\ &= \int (2\sec^2 u - 3)\, du = 2\tan u - 3u + c = 2x - 3\arctan x + c.\end{aligned}$$

Or

$$\begin{aligned}\frac{2x^2-1}{1+x^2} &\equiv \frac{2x^2-1+2-2}{1+x^2} = \frac{2(1+x^2)}{1+x^2} - \frac{3}{1+x^2} \\ &\equiv 2 - \frac{3}{1+x^2}.\end{aligned}$$

Polynomial division gives quotient 2 and remainder -3. Hence $\int \frac{2x^2-1}{1+x^2}\, dx = 2x - 3\arctan x + c$.

2 b)

Let $x = \tan u$, so $dx = \sec^2 u\, du$.

$$\begin{aligned}\int \frac{5x^2+9}{1+x^2}\, dx &= \int (5\tan^2 u + 9)\, du = \int \big(5(\sec^2 u - 1) + 9\big)\, du \\ &= \int (5\sec^2 u + 4)\, du = 5\tan u + 4u + c = 5x + 4\arctan x + c.\end{aligned}$$

Or

$$\begin{aligned}\frac{5x^2+9}{1+x^2} &\equiv \frac{5x^2+9+5-5}{1+x^2} = \frac{5(1+x^2)}{1+x^2} + \frac{4}{1+x^2} \\ &\equiv 5 + \frac{4}{1+x^2}.\end{aligned}$$

Polynomial division gives quotient 5 and remainder 4. Hence $\int \frac{5x^2+9}{1+x^2}\, dx = 5x + 4\arctan x + c$.

3 a)

Use integration by substitution. Let $\sqrt{x} = \tan u$. Differentiating gives $\frac{1}{2\sqrt{x}} = \sec^2 u \frac{du}{dx}$, so $dx = 2\tan u \sec^2 u\, du$.

$$\begin{aligned}\int \frac{1}{\sqrt{x}(1+x)}\, dx &= \int \frac{1}{\tan u(1+\tan^2 u)}\, 2\tan u \sec^2 u\, du \\ &= \int \frac{2\sec^2 u}{1+\tan^2 u}\, du = \int 2\, du \\ &= 2u + c = 2\arctan(\sqrt{x}) + c.\end{aligned}$$

3 b)

Use integration by substitution. Let $\sqrt{x} = u$, so $dx = 2u\, du$.

$$\begin{aligned}\int \frac{1}{\sqrt{x}(1+x)}\, dx &= \int \frac{1}{u(1+u^2)}\, 2u\, du = \int \frac{2}{1+u^2}\, du \\ &= 2\arctan u + c = 2\arctan(\sqrt{x}) + c.\end{aligned}$$

4)

Use integration by substitution. Let $x = \tan u$, so $dx = \sec^2 u\, du$.

$$\begin{aligned}\int \frac{1}{(1+x^2)^{3/2}}\, dx &= \int \frac{\sec^2 u}{(1+\tan^2 u)^{3/2}}\, du = \int \frac{\sec^2 u}{\sec^3 u}\, du \\ &= \int \cos u\, du = \sin u + c.\end{aligned}$$

Since $\tan u = \frac{x}{1} = \frac{\text{opp}}{\text{adj}}$, $\sin u = \frac{x}{\sqrt{1+x^2}}$. Therefore

$$\int \frac{1}{(1+x^2)^{3/2}}\, dx = \frac{x}{\sqrt{1+x^2}} + c.$$

5)

Use integration by substitution. Let $u = \sqrt{x^2-1}$. Then

$$\frac{du}{dx} = \frac{1}{2}\cdot 2x(x^2-1)^{-1/2} = \frac{x}{\sqrt{x^2-1}}, \qquad dx = \frac{\sqrt{x^2-1}}{x}\, du = \frac{u}{x}\, du.$$

Since $u^2 = x^2 - 1$, $x^2 = 1 + u^2$. Thus

$$\begin{aligned}\int \frac{1}{x\sqrt{x^2-1}}\, dx &= \int \frac{1}{xu}\cdot\frac{u}{x}\, du = \int \frac{1}{x^2}\, du = \int \frac{1}{1+u^2}\, du \\ &= \arctan u + c = \arctan\left(\sqrt{x^2-1}\right) + c.\end{aligned}$$

6 a)

$$\cos 2\theta \equiv 1 - 2\sin^2\theta \quad \Longrightarrow \quad \sin^2\theta \equiv \frac{1}{2} - \frac{1}{2}\cos 2\theta.$$

6 b)

Let $x = \sin u$, so $dx = \cos u\, du$.

$$\begin{aligned}\int \frac{x^2}{\sqrt{1-x^2}}\, dx &= \int \frac{\sin^2 u}{\sqrt{1-\sin^2 u}} \cos u\, du = \int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\, du \\ &= \int \sin^2 u\, du = \int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right) du \\ &= \frac{1}{2}u - \frac{1}{4}\sin 2u + c = \frac{1}{2}u - \frac{1}{2}\sin u \cos u + c \\ &= \frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1-x^2} + c.\end{aligned}$$

7 a)

Let $2x = \sin u$, so $x = \frac{1}{2}\sin u$ and $dx = \frac{1}{2}\cos u\, du$.

$$\begin{aligned}\int \frac{x^2}{\sqrt{1-4x^2}}\, dx &= \frac{1}{8}\int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\, du = \frac{1}{8}\int \sin^2 u\, du \\ &= \frac{1}{16}u - \frac{1}{32}\sin 2u + c = \frac{1}{16}u - \frac{1}{16}\sin u \cos u + c \\ &= \frac{1}{16}\arcsin(2x) - \frac{1}{8}x\sqrt{1-4x^2} + c.\end{aligned}$$

7 b)

Let $\frac{1}{2}x = \sin u$, so $x = 2\sin u$ and $dx = 2\cos u\, du$.

$$\begin{aligned}\int \frac{x^2}{\sqrt{4-x^2}}\, dx &= \frac{1}{2}\int \frac{x^2}{\sqrt{1-(x/2)^2}}\, dx = \frac{1}{2}\int \frac{4\sin^2 u}{\sqrt{\cos^2 u}}\, 2\cos u\, du \\ &= 4\int \sin^2 u\, du = 2u - \sin 2u + c \\ &= 2u - 2\sin u \cos u + c = 2\arcsin\left(\frac{x}{2}\right) - x\sqrt{1-\frac{x^2}{4}} + c.\end{aligned}$$

7 c)

Let $\frac{x}{\sqrt{2}} = \sin u$, so $x = \sqrt{2}\sin u$ and $dx = \sqrt{2}\cos u\, du$.

$$\begin{aligned}\int \frac{x^2}{\sqrt{4-2x^2}}\, dx &= \frac{1}{2}\int \frac{x^2}{\sqrt{1-(x/\sqrt{2})^2}}\, dx = \sqrt{2}\int \sin^2 u\, du \\ &= \frac{\sqrt{2}}{2}u - \frac{\sqrt{2}}{4}\sin 2u + c = \frac{\sqrt{2}}{2}u - \frac{\sqrt{2}}{2}\sin u \cos u + c \\ &= \frac{\sqrt{2}}{2}\arcsin\left(\frac{x}{\sqrt{2}}\right) - \frac{1}{2}x\sqrt{1-\frac{x^2}{2}} + c.\end{aligned}$$

8 a)

Let $x = \sin u$, so $dx = \cos u\, du$.

$$\begin{aligned}\int \frac{2x^2+3}{\sqrt{1-x^2}}\, dx &= \int (2\sin^2 u + 3)\, du = \int (4 - \cos 2u)\, du \\ &= 4u - \frac{1}{2}\sin 2u + c = 4u - \sin u \cos u + c \\ &= 4\arcsin x - x\sqrt{1-x^2} + c.\end{aligned}$$

8 b)

Let $3x = \sin u$, so $x = \frac{1}{3}\sin u$ and $dx = \frac{1}{3}\cos u\, du$.

$$\begin{aligned}\int \frac{3-x^2}{\sqrt{1-9x^2}}\,dx &= \int \left(1 - \frac{1}{27}\sin^2 u\right) du = \int \left(\frac{53}{54} + \frac{1}{54}\cos 2u\right) du \\ &= \frac{53}{54}u + \frac{1}{108}\sin 2u + c = \frac{53}{54}u + \frac{1}{54}\sin u\cos u + c \\ &= \frac{53}{54}\arcsin(3x) + \frac{1}{18}x\sqrt{1-9x^2} + c.\end{aligned}$$

8 c)

Let $\sqrt{3}x = \sin u$, so $x = \frac{1}{\sqrt{3}}\sin u$ and $dx = \frac{1}{\sqrt{3}}\cos u\, du$.

$$\begin{aligned}\int \frac{3x^2+1}{\sqrt{1-3x^2}}\,dx &= \int \left(\frac{1}{\sqrt{3}}\sin^2 u + \frac{1}{\sqrt{3}}\right) du \\ &= \int \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{6}\cos 2u\right) du = \frac{\sqrt{3}}{2}u - \frac{\sqrt{3}}{12}\sin 2u + c \\ &= \frac{\sqrt{3}}{2}u - \frac{\sqrt{3}}{6}\sin u\cos u + c \\ &= \frac{\sqrt{3}}{2}\arcsin(\sqrt{3}x) - \frac{1}{2}x\sqrt{1-3x^2} + c.\end{aligned}$$

9)

$$\int_{-1/4}^{1/4} \frac{8+8x^2}{\sqrt{1-4x^2}}\,dx = \int_{-1/4}^{1/4} \left(\frac{8}{\sqrt{1-(2x)^2}} + \frac{8x^2}{\sqrt{1-4x^2}}\right) dx.$$

For the second integral, let $2x = \sin u$. Then

$$\int \frac{8x^2}{\sqrt{1-4x^2}}\,dx = \int \sin^2 u\, du = \frac{1}{2}u - \frac{1}{2}\sin u\cos u = \frac{1}{2}\arcsin(2x) - x\sqrt{1-4x^2} + c.$$

Therefore

$$\begin{aligned}\int_{-1/4}^{1/4} \frac{8+8x^2}{\sqrt{1-4x^2}}\,dx &= \left[\frac{9}{2}\arcsin(2x) - x\sqrt{1-4x^2}\right]_{-1/4}^{1/4} \\ &= \left(\frac{3\pi}{4} - \frac{\sqrt{3}}{8}\right) - \left(-\frac{3\pi}{4} + \frac{\sqrt{3}}{8}\right) = \frac{3\pi}{2} - \frac{\sqrt{3}}{4}.\end{aligned}$$

10 a)

$$\begin{aligned}\frac{d}{dx}\left(3x\sqrt{1-9x^2} - \arccos(3x)\right) &= 3x\cdot\frac{1}{2}(-18x)(1-9x^2)^{-1/2} + 3(1-9x^2)^{1/2} + \frac{3}{\sqrt{1-9x^2}} \\ &= -\frac{27x^2}{\sqrt{1-9x^2}} + 3\sqrt{1-9x^2} + \frac{3}{\sqrt{1-9x^2}} \\ &= 3\sqrt{1-9x^2} + \frac{3-27x^2}{\sqrt{1-9x^2}} \\ &= 3\sqrt{1-9x^2} + 3\sqrt{1-9x^2} = 6\sqrt{1-9x^2}.\end{aligned}$$

So $k = 6$.

10 b)

$$\int_{1/6}^{\sqrt{3}/6} \sqrt{1-9x^2}\,dx = \frac{1}{6}\left[3x\sqrt{1-9x^2} - \arccos(3x)\right]_{1/6}^{\sqrt{3}/6}$$
$$= \frac{1}{6}\left(\left(\frac{\sqrt{3}}{4} - \frac{\pi}{6}\right) - \left(\frac{\sqrt{3}}{4} - \frac{\pi}{3}\right)\right) = \frac{\pi}{36}.$$

11 a)

$$\frac{d}{dx}\left(2(x-5)\sqrt{11+10x-x^2}\right) + \frac{d}{dx}\left(72\arcsin\left(\frac{x-5}{6}\right)\right)$$
$$= -\frac{2(x-5)^2}{\sqrt{11+10x-x^2}} + 2\sqrt{11+10x-x^2} + \frac{72}{\sqrt{36-(x-5)^2}}$$
$$= 2\sqrt{11+10x-x^2} + \frac{72-2(x-5)^2}{\sqrt{11+10x-x^2}}$$
$$= 2\sqrt{11+10x-x^2} + 2\sqrt{11+10x-x^2}$$
$$= 4\sqrt{11+10x-x^2}.$$

So $k = 4$.

11 b)

$$\int_5^8 \sqrt{11+10x-x^2}\,dx = \left[\frac{1}{2}(x-5)\sqrt{11+10x-x^2} + 18\arcsin\left(\frac{x-5}{6}\right)\right]_5^8$$
$$= \left(\frac{1}{2}\cdot 3\cdot 3\sqrt{3} + 18\arcsin\frac{1}{2}\right) - 0$$
$$= \frac{9\sqrt{3}}{2} + 3\pi.$$

12 a)

$$19+18x-x^2 \equiv 19-(x^2-18x)$$
$$\equiv 19-\left((x-9)^2-81\right) = 100-(x-9)^2.$$

So $p = 100$ and $q = -9$.

12 b)

Let $\frac{x-9}{10} = \sin u$, so $x = 10\sin u + 9$ and $dx = 10\cos u\,du$.

$$\int \sqrt{19+18x-x^2}\,dx = 10\int \sqrt{1-\left(\frac{x-9}{10}\right)^2}\,dx = 100\int \cos^2 u\,du$$
$$= 25\sin 2u + 50u + c = 50\sin u\cos u + 50u + c$$
$$= \frac{1}{2}(x-9)\sqrt{100-(x-9)^2} + 50\arcsin\left(\frac{x-9}{10}\right) + c$$
$$= \frac{1}{2}(x-9)\sqrt{19+19x-x^2} + 50\arcsin\left(\frac{x-9}{10}\right) + c.$$

13)

Complete the square: $3+2x-x^2 = 4-(x-1)^2$. Let $\frac{x-1}{2} = \sin u$, so $x = 2\sin u+1$ and $dx = 2\cos u\,du$.

$$\begin{aligned}\int \sqrt{3+2x-x^2}\,dx &= 4\int \cos^2 u\,du = \sin 2u + 2u + c \\ &= \frac{1}{2}(x-1)\sqrt{4-(x-1)^2} + 2\arcsin\left(\frac{x-1}{2}\right) + c \\ &= \frac{1}{2}(x-1)\sqrt{3+2x-x^2} + 2\arcsin\left(\frac{x-1}{2}\right) + c.\end{aligned}$$

14 a)

$$\begin{aligned}\sin 3A \equiv \sin(2A+A) &= \sin 2A\cos A + \cos 2A\sin A \\ &\equiv 2\sin A\cos^2 A + (1-2\sin^2 A)\sin A \\ &\equiv 2\sin A(1-\sin^2 A) + \sin A - 2\sin^3 A = 3\sin A - 4\sin^3 A.\end{aligned}$$

Alternatively, $(\cos A + i\sin A)^3 \equiv \cos 3A + i\sin 3A$. Equating imaginary parts gives $\sin 3A \equiv 3\cos^2 A\sin A - \sin^3 A = 3\sin A - 4\sin^3 A$.

14 b)

$$\begin{aligned}\cos 3A \equiv \cos(2A+A) &= \cos 2A\cos A - \sin 2A\sin A \\ &\equiv (2\cos^2 A - 1)\cos A - 2\sin^2 A\cos A \\ &\equiv 2\cos^3 A - \cos A - 2(1-\cos^2 A)\cos A = 4\cos^3 A - 3\cos A.\end{aligned}$$

Alternatively, $(\cos A + i\sin A)^3 \equiv \cos 3A + i\sin 3A$. Equating real parts gives $\cos 3A \equiv \cos^3 A - 3\cos A\sin^2 A = 4\cos^3 A - 3\cos A$.

14 c)

Let $x = \sin u$, so $dx = \cos u\,du$. Using $\sin 3u = 3\sin u - 4\sin^3 u$,

$$\begin{aligned}I = \int \frac{6x^3-4x+1}{\sqrt{1-x^2}}\,dx &= \int (6\sin^3 u - 4\sin u + 1)\,du \\ &= \int \left(\frac{1}{2}\sin u - \frac{3}{2}\sin 3u + 1\right)du \\ &= -\frac{1}{2}\cos u + \frac{1}{2}\cos 3u + u + c \\ &= -2\cos u + 2\cos^3 u + u + c \\ &= -2\sqrt{1-x^2} + 2(1-x^2)^{3/2} + \arcsin x + c.\end{aligned}$$

Alternatively, write $\sin^3 u = \sin u(1-\cos^2 u)$ and integrate $2\sin u - 6\sin u\cos^2 u + 1$ to obtain the same result.

15 a)

$$\frac{d}{dx}\left(\arctan\sqrt{\frac{x}{3}}\right) = \frac{1}{2\sqrt{3x}\,(1+\frac{x}{3})} = \frac{\sqrt{3}}{2\sqrt{x}(3+x)}.$$

15 b)

Since $\frac{d}{dx}\arctan\sqrt{x/3} = \frac{\sqrt{3}}{2}\frac{\sqrt{x}}{3x+x^2}$,

$$\int_1^3 \frac{\sqrt{x}}{3x+x^2}\,dx = \frac{2\sqrt{3}}{3}\left[\arctan\sqrt{\frac{x}{3}}\right]_1^3 = \frac{2\sqrt{3}}{3}\left(\frac{\pi}{4}-\frac{\pi}{6}\right) = \frac{\pi}{6\sqrt{3}}.$$

16)

Let $u^2 = x-3$, where $u>0$. Then $dx = 2u\,du$ and $x+6 = u^2+9$.

$$\begin{aligned}\int \frac{\sqrt{x-3}}{x+6}\,dx &= \int \frac{2u^2}{u^2+9}\,du = \int\left(2-\frac{18}{u^2+9}\right)du\\ &= 2u - 6\arctan\left(\frac{u}{3}\right)+c\\ &= 2\sqrt{x-3} - 6\arctan\left(\frac{\sqrt{x-3}}{3}\right)+c.\end{aligned}$$

Polynomial division gives the same decomposition, $\frac{2u^2}{9+u^2} = 2-\frac{18}{9+u^2}$.

17)

Let $u^2 = x+1$, where $u>0$. Then $dx = 2u\,du$ and $x+10 = u^2+9$.

$$\begin{aligned}\int \frac{x}{\sqrt{x+1}(x+10)}\,dx &= \int \frac{2u^2-2}{u^2+9}\,du = \int\left(2-\frac{20}{u^2+9}\right)du\\ &= 2u - \frac{20}{3}\arctan\left(\frac{u}{3}\right)+c\\ &= 2\sqrt{x+1} - \frac{20}{3}\arctan\left(\frac{\sqrt{x+1}}{3}\right)+c.\end{aligned}$$

Polynomial division gives the same decomposition, $\frac{2u^2-2}{u^2+9} = 2-\frac{20}{u^2+9}$.

18)

For the shaded area, let $u^2 = x+4$, where $u>0$. Then $dx = 2u\,du$ and $x+7 = u^2+3$. When $x=-3$, $u=1$; when $x=-1$, $u=\sqrt{3}$.

$$\begin{aligned}\int_{-3}^{-1} \frac{\sqrt{x+4}}{x+7}\,dx &= \int_1^{\sqrt{3}} \frac{2u^2}{u^2+3}\,du = \int_1^{\sqrt{3}}\left(2-\frac{6}{u^2+3}\right)du\\ &= \left[2u - 2\sqrt{3}\arctan\left(\frac{u}{\sqrt{3}}\right)\right]_1^{\sqrt{3}}\\ &= 2\sqrt{3}-2+\left(\frac{\sqrt{3}}{3}-\frac{\sqrt{3}}{2}\right)\pi = 2\sqrt{3}-2+\frac{\pi}{2\sqrt{3}}\ \text{units}^2.\end{aligned}$$

19 a)

Use integration by parts with $u=\arcsin x$, $dv = dx$, $du = \frac{dx}{\sqrt{1-x^2}}$ and $v = x$.

$$\int \arcsin x\,dx = x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\,dx,$$

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\frac{1}{2}\int w^{-1/2}\,dw = -\sqrt{1-x^2}+c, \qquad w = 1-x^2,; dw = -2x\,dx.$$

Hence $\int \arcsin x\,dx = x\arcsin x + \sqrt{1-x^2}+c$.

19 b)

Using integration by parts with $u = \arccos x$, $dv = dx$, $du = -\frac{dx}{\sqrt{1-x^2}}$ and $v = x$ gives

$$\int \arccos x \, dx = x \arccos x + \int \frac{x}{\sqrt{1-x^2}} \, dx = x \arccos x - \sqrt{1-x^2} + c.$$

19 c)

Using integration by parts with $u = \arctan x$, $dv = dx$, $du = \frac{dx}{1+x^2}$ and $v = x$ gives

$$\int \arctan x \, dx = x \arctan x - \int \frac{x}{1+x^2} \, dx = x \arctan x - \frac{1}{2} \ln |1+x^2| + c.$$

20)

Use integration by parts with $u = \arctan x$, $dv = 8\,dx$, $du = \frac{dx}{1+x^2}$ and $v = 8x$.

$$\begin{aligned} \int_0^1 8 \arctan x \, dx &= \Big[8x \arctan x + 4 \ln |1+x^2| \Big]_0^1 \\ &= 2\pi + 4 \ln 2 = 2\pi + \ln 16. \end{aligned}$$

So the printed worked solutions give $p = 16$.

21 a)

Integration by parts gives

$$\int \arcsin(2x) \, dx = x \arcsin(2x) - \int \frac{2x}{\sqrt{1-4x^2}} \, dx = x \arcsin(2x) + \frac{1}{2}\sqrt{1-4x^2} + c,$$

where the remaining integral uses $w = 1 - 4x^2$, $dw = -8x\,dx$.

21 b)

Integration by parts gives

$$\int \arccos(3x) \, dx = x \arccos(3x) + \int \frac{3x}{\sqrt{1-9x^2}} \, dx = x \arccos(3x) - \frac{1}{3}\sqrt{1-9x^2} + c,$$

where the remaining integral uses $w = 1 - 9x^2$, $dw = -18x\,dx$.

21 c)

Integration by parts gives

$$\int \arctan(4x) \, dx = x \arctan(4x) - \int \frac{4x}{1+16x^2} \, dx = x \arctan(4x) - \frac{1}{8} \ln |1+16x^2| + c,$$

where the remaining integral uses $w = 1 + 16x^2$, $dw = 32x\,dx$.

22 a)

By reversing the chain rule and using the result of Question 19(a),

$$\begin{aligned} \int 8x \arcsin(x^2) \, dx &= 4 \left(x^2 \arcsin(x^2) + \sqrt{1-(x^2)^2} \right) + c \\ &= 4x^2 \arcsin(x^2) + 4\sqrt{1-x^4} + c. \end{aligned}$$

22 b)

Using $\int \arccos u\,du = u\arccos u - \sqrt{1-u^2} + c$ and reversing the chain rule,

$$\int e^{2x}\arccos(e^{2x})\,dx = \frac{1}{2}e^{2x}\arccos(e^{2x}) - \frac{1}{2}\sqrt{1-e^{4x}} + c.$$

22 c)

Using $\int \arctan u\,du = u\arctan u - \frac{1}{2}\ln|1+u^2| + c$ with $u = \cos x$ gives

$$\int \sin x\arctan(\cos x)\,dx = -\cos x\arctan(\cos x) + \frac{1}{2}\ln|1+\cos^2 x| + c.$$

22 d)

Using $\int \arcsin u\,du = u\arcsin u + \sqrt{1-u^2} + c$ with $u = \tan x$ gives

$$\int \sec^2 x\arcsin(\tan x)\,dx = \tan x\arcsin(\tan x) + \sqrt{1-\tan^2 x} + c.$$

22 e)

Using $\int \arctan u\,du = u\arctan u - \frac{1}{2}\ln|1+u^2| + c$ and reversing the chain rule,

$$\int 3^x\arctan(3^x)\,dx = \frac{1}{\ln 3}\left(3^x\arctan(3^x) - \frac{1}{2}\ln|1+3^{2x}|\right) + c.$$

23 a)

Use integration by parts with $u = \arcsin x$ and $dv = x\,dx$:

$$\int x\arcsin x\,dx = \frac{1}{2}x^2\arcsin x - \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}}\,dx.$$

For the remaining integral let $x = \sin u$, so $\int \frac{x^2}{\sqrt{1-x^2}}\,dx = \int \sin^2 u\,du = \frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1-x^2} + c$. Hence

$$\int x\arcsin x\,dx = \frac{1}{2}x^2\arcsin x - \frac{1}{4}\arcsin x + \frac{1}{4}x\sqrt{1-x^2} + c.$$

23 b)

Use integration by parts with $u = \arccos x$ and $dv = x^2\,dx$:

$$\int x^2\arccos x\,dx = \frac{1}{3}x^3\arccos x + \frac{1}{3}\int \frac{x^3}{\sqrt{1-x^2}}\,dx.$$

Let $w = 1-x^2$, so $x^2 = 1-w$. Then

$$\begin{aligned}\int \frac{x^3}{\sqrt{1-x^2}}\,dx &= -\frac{1}{2}\int (1-w)w^{-1/2}\,dw \\ &= \frac{1}{3}w^{3/2} - w^{1/2} + c = \frac{1}{3}(1-x^2)^{3/2} - \sqrt{1-x^2} + c.\end{aligned}$$

Therefore $\int x^2\arccos x\,dx = \frac{1}{3}x^3\arccos x + \frac{1}{9}(1-x^2)^{3/2} - \frac{1}{3}\sqrt{1-x^2} + c.$

23 c)

Use integration by parts with $u = \arctan x$ and $dv = x^3\,dx$:

$$\int x^3 \arctan x\,dx = \frac{1}{4}x^4 \arctan x - \frac{1}{4}\int \frac{x^4}{1+x^2}\,dx.$$

Polynomial division gives $\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}$. Thus

$$\int x^3 \arctan x\,dx = \frac{1}{4}x^4 \arctan x - \frac{1}{12}x^3 + \frac{1}{4}x - \frac{1}{4}\arctan x + c.$$

24 a)

$$\begin{aligned}\frac{10x^2}{1+x^2} &\equiv \frac{10x^2+10-10}{1+x^2} = \frac{10(1+x^2)}{1+x^2} - \frac{10}{1+x^2} \\ &\equiv 10 - \frac{10}{1+x^2}.\end{aligned}$$

Or using polynomial division, the quotient is 10 and the remainder is -10, giving the same identity.

24 b)

Use integration by parts with $u = \arctan x$ and $dv = 20x\,dx$, so $du = \frac{dx}{1+x^2}$ and $v = 10x^2$.

$$\begin{aligned}\int 20x \arctan x\,dx &= 10x^2 \arctan x - \int \frac{10x^2}{1+x^2}\,dx \\ &= 10x^2 \arctan x - \int \left(10 - \frac{10}{1+x^2}\right) dx \\ &= 10x^2 \arctan x - 10x + 10\arctan x + c.\end{aligned}$$

25 a)

$$\begin{aligned}\frac{15x^2}{4+x^2} &\equiv \frac{15x^2+60-60}{4+x^2} = \frac{15(4+x^2)}{4+x^2} - \frac{60}{4+x^2} \\ &\equiv 15 - \frac{60}{4+x^2}.\end{aligned}$$

Or using polynomial division, the quotient is 15 and the remainder is -60, giving the same identity.

25 b)

Use integration by parts with $u = \arctan(x/2)$ and $dv = 30x\,dx$, so $du = \frac{2}{4+x^2}dx$ and $v = 15x^2$.

$$\begin{aligned}\int 30x \arctan\left(\frac{x}{2}\right) dx &= 15x^2 \arctan\left(\frac{x}{2}\right) - \int \frac{30x^2}{4+x^2}\,dx \\ &= 15x^2 \arctan\left(\frac{x}{2}\right) - \int \left(30 - \frac{120}{4+x^2}\right) dx \\ &= 15x^2 \arctan\left(\frac{x}{2}\right) - 30x + 60\arctan\left(\frac{x}{2}\right) + c.\end{aligned}$$

26)

Let $u = x^2$, so $du = 2x\,dx$. The limits remain $u = 0$ to $u = 1$.

$$\int_0^1 6x \arcsin(x^2)\,dx = 3\int_0^1 \arcsin u\,du.$$

By integration by parts, $\int \arcsin u\,du = u\arcsin u + \sqrt{1-u^2} + c$. Hence

$$\begin{aligned} 3\int_0^1 \arcsin u\,du &= 3\left[u\arcsin u + \sqrt{1-u^2}\right]_0^1 \\ &= 3\left(\frac{\pi}{2} - 1\right) = \left(\frac{3\pi}{2} - 3\right) \text{ units}^2. \end{aligned}$$

27 a)

Integration by parts with $u = \arcsin x$ and $dv = \frac{dx}{\sqrt{1-x^2}}$ gives

$$I = \int \frac{\arcsin x}{\sqrt{1-x^2}}\,dx = (\arcsin x)^2 - I,$$

so $I = \frac{1}{2}(\arcsin x)^2 + c$. Alternatively, let $u = \arcsin x$, so $du = \frac{dx}{\sqrt{1-x^2}}$; then $I = \int u\,du = \frac{1}{2}u^2 + c = \frac{1}{2}(\arcsin x)^2 + c$.

27 b)

Integration by parts with $u = \arccos x$ and $dv = \frac{dx}{\sqrt{1-x^2}}$ gives

$$I = \int \frac{\arccos x}{\sqrt{1-x^2}}\,dx = -(\arccos x)^2 - I,$$

so $I = -\frac{1}{2}(\arccos x)^2 + c$. Alternatively, let $u = \arccos x$, so $du = -\frac{dx}{\sqrt{1-x^2}}$; then $I = -\int u\,du = -\frac{1}{2}u^2 + c$.

27 c)

Integration by parts with $u = \arctan x$ and $dv = \frac{dx}{1+x^2}$ gives

$$I = \int \frac{\arctan x}{1+x^2}\,dx = (\arctan x)^2 - I,$$

so $I = \frac{1}{2}(\arctan x)^2 + c$. Alternatively, let $u = \arctan x$, so $du = \frac{dx}{1+x^2}$; then $I = \int u\,du = \frac{1}{2}u^2 + c$.

28)

Let $I = \int_{1/2}^{1/\sqrt{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,dx$. Integration by parts with $u = (\arcsin x)^2$ and $dv = \frac{dx}{\sqrt{1-x^2}}$ gives

$$\begin{aligned} I &= \left[(\arcsin x)^3\right]_{1/2}^{1/\sqrt{2}} - 2I, \\ I &= \frac{1}{3}\left[(\arcsin x)^3\right]_{1/2}^{1/\sqrt{2}} \\ &= \frac{1}{3}\left(\left(\frac{\pi}{4}\right)^3 - \left(\frac{\pi}{6}\right)^3\right) = \frac{19\pi^3}{5184}. \end{aligned}$$

Thus $k = \frac{19}{5184}$. Alternatively, set $u = \arcsin x$, so the limits become $\pi/6$ and $\pi/4$ and $I = \int_{\pi/6}^{\pi/4} u^2\,du = \frac{1}{3}[u^3]_{\pi/6}^{\pi/4} = \frac{19\pi^3}{5184}$.

29)

First, integrate $\ln(1+x^2)$ by parts:

$$\begin{aligned}\int \ln(1+x^2)\,dx &= x\ln(1+x^2) - \int \frac{2x^2}{1+x^2}\,dx \\ &= x\ln(1+x^2) - \int \left(2 - \frac{2}{1+x^2}\right) dx \\ &= x\ln(1+x^2) - 2x + 2\arctan x + c.\end{aligned}$$

Now use integration by parts on the original integral with $u = \arctan x$ and $dv = \ln(1+x^2)\,dx$:

$$\begin{aligned}\int \arctan x \ln(1+x^2)\,dx = (x\ln(1+x^2) - 2x + 2\arctan x)\arctan x \\ - \int \frac{x\ln(1+x^2) - 2x + 2\arctan x}{1+x^2}\,dx.\end{aligned}$$

For the remaining integrals, use $u = 1+x^2$ in the first two and $u = \arctan x$ in the third:

$$\begin{aligned}\int \frac{x\ln(1+x^2)}{1+x^2}\,dx &= \frac{1}{4}(\ln(1+x^2))^2 + c, \\ \int \frac{2x}{1+x^2}\,dx &= \ln(1+x^2) + c, \\ \int \frac{2\arctan x}{1+x^2}\,dx &= (\arctan x)^2 + c.\end{aligned}$$

Therefore

$$\begin{aligned}\int \arctan x \ln(1+x^2)\,dx = (x\ln(1+x^2) - 2x + 2\arctan x)\arctan x \\ - \frac{1}{4}(\ln(1+x^2))^2 + \ln(1+x^2) - (\arctan x)^2 + c.\end{aligned}$$

30 a)

$$\int \operatorname{arccosec} x\,dx = \int 1 \cdot \operatorname{arccosec} x\,dx.$$

To use integration by parts, first find the derivative of $\operatorname{arccosec} x$. Let $y = \operatorname{arccosec} x$, so $\operatorname{cosec} y = x$. Differentiating,

$$-\operatorname{cosec} y \cot y \frac{dy}{dx} = 1, \qquad \frac{dy}{dx} = -\frac{1}{\operatorname{cosec} y\sqrt{\operatorname{cosec}^2 y - 1}} = -\frac{1}{x\sqrt{x^2-1}}.$$

Now use integration by parts with $u = \operatorname{arccosec} x$ and $dv = dx$:

$$\int \operatorname{arccosec} x\,dx = x\operatorname{arccosec} x + \int \frac{1}{\sqrt{x^2-1}}\,dx = x\operatorname{arccosec} x + \operatorname{arcosh} x + c.$$

30 b)

$$\int \operatorname{arcsec} x\,dx = \int 1 \cdot \operatorname{arcsec} x\,dx.$$

The printed solutions state that the derivative of $\operatorname{arccosec} x$ is needed here. Let $y = \operatorname{arcsec} x$, so $\sec y = x$. Differentiating,

$$\sec y \tan y \frac{dy}{dx} = 1, \qquad \frac{dy}{dx} = \frac{1}{\sec y\sqrt{\sec^2 y - 1}} = \frac{1}{x\sqrt{x^2-1}}.$$

Thus integration by parts gives

$$\int \operatorname{arcsec} x\,dx = x\operatorname{arcsec} x - \int \frac{1}{\sqrt{x^2-1}}\,dx = x\operatorname{arcsec} x - \operatorname{arcosh} x + c.$$

30 c)

$$\int \operatorname{arccot} x\,dx = \int 1 \cdot \operatorname{arccot} x\,dx.$$

The printed solutions state that the derivative of $\operatorname{arccosec} x$ is needed here. Let $y = \operatorname{arccot} x$, so $\cot y = x$. Differentiating,

$$-\operatorname{cosec}^2 y \frac{dy}{dx} = 1, \qquad \frac{dy}{dx} = -\frac{1}{\operatorname{cosec}^2 y} = -\frac{1}{x^2+1}.$$

Integration by parts then gives

$$\int \operatorname{arccot} x\,dx = x \operatorname{arccot} x + \int \frac{x}{x^2+1}\,dx = x \operatorname{arccot} x + \frac{1}{2}\ln|x^2+1| + c.$$

31)

Let $y = \operatorname{arccosec}(2x)$, so $\operatorname{cosec} y = 2x$. Differentiating,

$$-\operatorname{cosec} y \cot y \frac{dy}{dx} = 2, \qquad \frac{dy}{dx} = -\frac{1}{x\sqrt{4x^2-1}}.$$

Use integration by parts with $u = \operatorname{arccosec}(2x)$ and $dv = dx$:

$$\begin{aligned}\int \operatorname{arccosec}(2x)\,dx &= x \operatorname{arccosec}(2x) + \int \frac{1}{\sqrt{4x^2-1}}\,dx \\ &= x \operatorname{arccosec}(2x) + \frac{1}{2}\operatorname{arcosh}(2x) + c.\end{aligned}$$

Therefore

$$\begin{aligned}\int_{1/\sqrt{2}}^{1} \operatorname{arccosec}(2x)\,dx &= \left[x \operatorname{arccosec}(2x) + \frac{1}{2}\operatorname{arcosh}(2x)\right]_{1/\sqrt{2}}^{1} \\ &= \left(\frac{\pi}{6} + \frac{1}{2}\ln(2+\sqrt{3})\right) - \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{2}\ln(1+\sqrt{2})\right) \\ &= \left(\frac{1}{2}\ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right) + \frac{\pi}{6} - \frac{\pi}{4\sqrt{2}}\right) \text{ units}^2.\end{aligned}$$